

Preprocessing Steps

1. Data Collection

- Gather housing datasets (CSV/Excel) containing sale prices, property features, and location details.
- Integrate external sources (government open data, real estate APIs).

2. Data Cleaning

- Handle missing values (impute or remove).
- Remove duplicates and inconsistent records.
- Normalize categorical fields (e.g., property type, neighborhood).

3. Data Transformation

- Convert dates into usable formats (month/year).
- Create derived variables (price per square foot, age of property).
- Encode categorical variables for analysis.

4. Data Integration

- Merge multiple datasets (features, sale prices, demographics).
- Ensure consistent schema across sources.

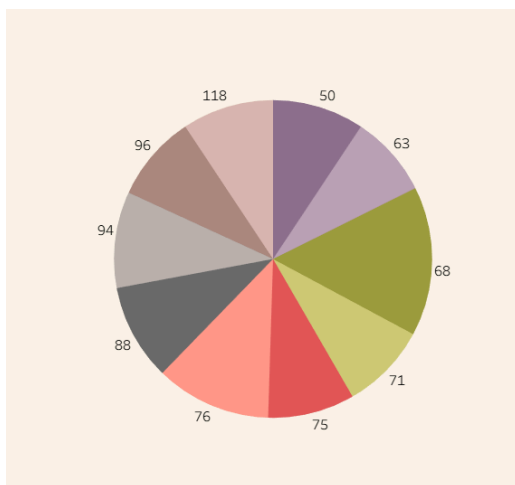
5. Data Validation

- Check for outliers in sale prices.
- Verify accuracy of location and property features.

Business Questions with Visualizations

Which property types dominate the market?

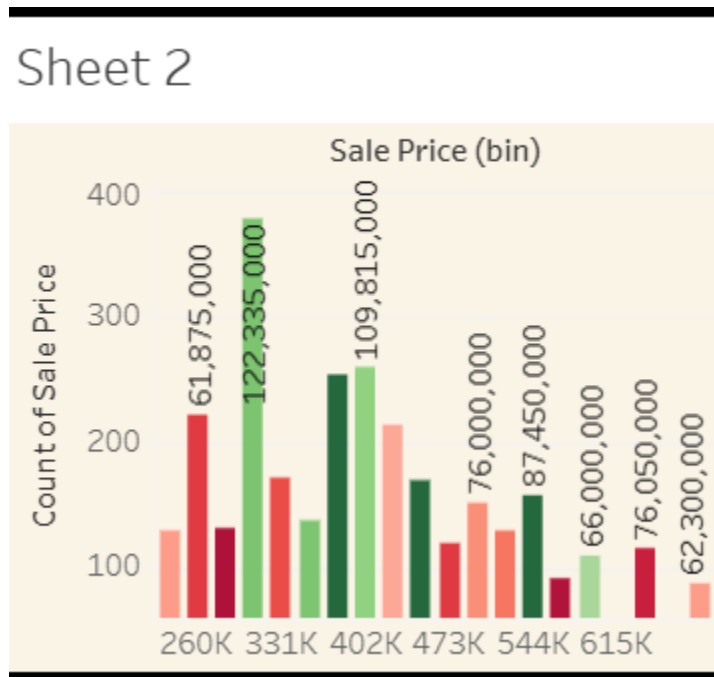
- Visualization:



- Insight: Shows market composition (apartments, villas, plots).

What is the average sale price by property type?

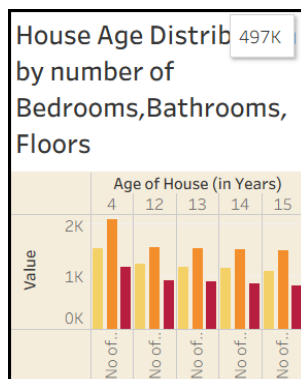
- Visualization:



- Insight: Compare average prices across property categories.

How do housing prices vary across neighborhoods?

Visualization:



- Insight: Compare sale prices across multiple neighborhoods side-by-side.

What is the distribution of sale prices?

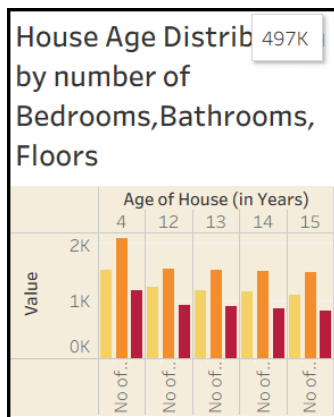
- Visualization:

Area of the House from Basement (in Sqft)	38,643,798
Count of Transformed_ Housing_Data2. csv	21,609
Sum of Sale Price	11,055,565,423

-
- Insight: Provides a clear numeric overview of pricing spread.

Which features most influence sale price?

- Visualization:



- Insight: Identifies correlations between property features and pricing.